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Review of hormonal replacement therapy options for the treatments of menopausal symptoms

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ABSTRACT

Menopause is a natural part of the aging process that every woman experiences at some point in life. Menopausal symptoms have a negative impact on the quality of life. Common menopausal symptoms include vasomotor symptoms, mood swings, concentration issues, vaginal dryness, atrophy of secondary sexual traits, libido loss, musculoskeletal discomfort, osteopenia, and osteoporosis. The most effective treatment for the relief of menopausal symptoms is estrogen, with or without. Hormone replacement therapy (HRT) is most beneficial before 60 years of age or within 10 years of menopause. Other menopause-related symptoms including mood swings, sleep disturbance, sexual dysfunction, and myalgia may improve with HRT. HRT is also effective in preventing bone loss associated with menopause and in reducing the incidence of all osteoporosis-related fractures, including those of the vertebrae and hip.

Keywords: Menopause; Hot Flashes; Estrogen Replacement Therapy; Clonidine; Selective Serotonin Reuptake Inhibitors

Introduction

Menopause is caused by changes in the hypothalamus and pituitary hormones, as well as the inability of the ovary to respond to pituitary hormones and the loss of ovarian follicles, as a result of apoptosis or programmed cell death. Menopause can also result from surgery or early ovarian insufficiency caused by genetic, immunological, metabolic, or infectious factors. Women have approximately 7,000,000 oocytes at birth but release only 500 oocytes throughout the course of their reproductive lives. Therefore, a decrease in the number of follicles in the ovary or an increase in the rate of follicle loss, may cause ovarian insufficiency. Oocyte apoptosis may also occur as a consequence of aberrant pairing during meiosis. Surgical removal of the ovaries usually results in severe endometriosis, cervical cancer, ovarian cancer, radical hysterectomy, or surgical menopause. Chemotherapy (cyclophosphamide and anthracycline) and/or radiation treatment can also cause this condition [1].

At a median age of 51.4 years, the menstrual cycle spontaneously ends, a phenomenon known as menopause. Vasomotor symptoms (VMS) are frequent and are treated with estrogen replacement therapy. It is important to distinguish between the strong ethinyl estradiol produced in combination with oral contraceptive regimens and hormone replacement therapy (HRT), in which estrogen is naturally comparable to that produced by the ovaries. If a woman still has a uterus, she must be administered

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progestogen or micronized progesterone to prevent endometrial hyperplasia and cancer. Micronized estradiol, estradiol valerate, estrone, estriol, or conjugated equine estrogens (CEEs) are all oral forms of estradiol [2].

Most women undergo certain physiological changes throughout their biopsychosocial stages of productive life, which can be affected by a variety of social, psychological, cultural, and ethnic variables. There is growing evidence that lifestyle factors (diet, exercise, smoking, and reproductive history), socioeconomic status, body mass index, mood, climate, and cognition (including attributing menopausal symptoms to menopause and attitudes and beliefs about it) may account for cultural differences in menopausal symptom reporting [3].

Common menopausal symptoms include VMS, mood swings, concentration issues, vaginal dryness, atrophy of secondary sexual traits, libido loss, musculoskeletal discomfort, osteopenia, and osteoporosis. VMS frequency varies according to geographic location. During the premenopausal stage, the distressing symptoms last for 2–3 years. It commonly occurs in iatrogenic menopause and early ovarian failure [1].

Up to 70%–80% of women have VMS (hot flashes and/or night sweats), which are typical after menopause. When taken locally, HRT is the most effective treatment for VMS, as well as for genitourinary syndrome of menopause (GSM), bone loss, and fracture risk. Women can use various forms, formulations, and methods of HRT, each of which may have unique dangers and advantages [4]. More than 85% of menopausal women suffer from these symptoms, which typically endure for 5–7 years, but can linger for 15 years or more [5]. Numerous unsettling symptoms may arise as ovarian hormone function declines [6].

Menopause symptoms have a negative impact on the quality of life. No therapy is more effective than HRT. However, the risk-benefit ratio has long been a topic of discussion. Its use was reduced by 80% after publication of the Women's Health Initiative report. Independent studies have shown that there is no justification for the persistently unfavorable perception of the use of HRT to treat symptoms [2].

Every woman must be evaluated based on her worries and physical results are specific to her. Although alternative methods may be beneficial, estrogen is the most effective treatment for women who wish to reduce their VMS. When a woman has VMS along with other illnesses (such as osteopenia or GSM), HRT becomes a more valuable treatment option. For healthy women within 10 years of menopause, who may be at an increased risk of chronic illnesses, HRT is a suitable alternative, even if there are no noticeable VMS [7].

Lifestyle, food, exercise, smoking, and alcohol usage should all be considered while making the choice to utilize HRT. Women with moderate-to-severe menopausal symptoms who do not have any contraindications appear to be able to safely undergo HRT. Using estrogen with or without progestogen is the most efficient way to alleviate menopausal symptoms such as hot flashes, night sweats, and urogenital atrophy. In the 10 years following menopause or before the age of 60 years, HRT is most effective. HRT may also alleviate other menopausal symptoms, such as mood swings, sleep issues, sexual dysfunction, and myalgia [3].

Menopause

Every woman eventually experiences menopause, which is a normal aspect of aging. Every woman experiences this shift at a different age. Most women experience menopause between the ages of 45 and 55 years; however, one in 100 may likely have menopausal symptoms before the age of 40 years, a condition known as "premature menopause." Women who smoke or have long-term medical conditions are more likely to experience menopause early in life. Life expectancy and future health status are affected by the age at which women approach menopause. Women who undergo menopause later in life (at age 55 years or older) tend to live noticeably longer than those who do so before the age of 40 years [8].

Women experience a number of problems, including vasomotor instability, hot flashes, night sweats, mood swings, and sleep disturbances, when their ovaries stop functioning and their estrogen levels decrease. Long-term estrogen insufficiency can also result in osteoporosis and cardiovascular problems, which are potentially fatal. Between 60%–80% of postmenopausal women experience unpleasant VMS [9].

HRT, approved by the Federal Drug Administration (FDA), replenishes ovarian hormones that are depleted during the natural menopausal transition, relieving symptoms such as severe vasomotor menopausal symptoms and preventing osteoporosis. There is substantial evidence that HRT reduces the incidence of osteoporosis, the risk of osteoporosis-related fractures, as well as the risk of cardiovascular disease (CVD) in some countries, thereby having a positive impact on life expectancy [10].

Indications Hormone Replacement Therapy

The primary indication for HRT is the alleviation of problematic VMS, which negatively affects quality of life. Urogenital atrophy and VMS are best treated with HRT. Other menopausal symptoms for which estrogen may help include mood changes, joint or muscle discomfort, sleep disturbances, and sexual dysfunction. However, HRT is not primarily used to address poor mood and libido. The choice to begin and maintain HRT is made depending on the type and intensity of menopausal symptoms, their effect on function, and the woman's unique health profile. Most clinical guidelines suggest beginning low-dose treatment; however, the dose is often titrated to alleviate side effects and symptoms, and there are no set rules on how long it should take.

Although approximately 40% of women aged 60–65 years may continue to experience VMS, the dangers associated with therapy may increase with increased usage. It is advised to follow up once a year to assess overall health and determine whether further HRT is necessary. Although it is typically not advised to begin HRT after the age of 60 years or more than 10 years after menopause, continuing current medication may be an option if bothersome symptoms persist. Sexual dysfunction may be a greater concern in young women undergoing menopause, and their symptoms may be more severe. According to current standards, women with early menopause (less than 45 years of age) or premature ovarian insufficiency (40 years of age) should continue receiving HRT until they reach the usual menopausal age, which is approximately 50 years. As with other post-menopausal women, continued usage after this point should be supported by the same risk-benefit analyses at regular intervals [11].

Hormonal Replacement Therapy Option

Estrogen therapy

Postmenopausal women who have had a hysterectomy are treated with estrogen alone. Human estrogens (17β-estradiol [E2], estrone [E1], and estriol [E3]), animal-derived estrogens (oCEE), and synthetic estrogens (ethinyl estradiol [EE]) are examples of estrogen formulations. E2, the most physiologically active and principal estrogen generated by the ovaries, is the only human estrogen formulation that has received FDA approval. A blend of natural estrogens, including those found in the urine of pregnant mares, make up the oCEE. The gastrointestinal system, skin, and mucous membranes absorb estrogen. Some formulations can be taken orally, transdermally, as sprays, gels, or topical emulsions, vaginally, in conjunction with progesterone, or as a tissue-specific estrogen complex (TSEC). The most popular estrogen treatment method is the oral route. Estrone is the main hormone in the bloodstream after estradiol is transformed during first-pass metabolism. First-pass metabolism of oral estrogens causes risks, including hypertriglyceridemia, increased coagulation factor and inflammatory marker production, increased risk of venous thromboembolism (VTE), and gallstones [4].

Oral estrogens increase sex hormone-binding globulin, triglyceride, and C-reactive protein levels owing to first-pass hepatic metabolism, and transdermal delivery prevents these effects [12]. However, depending on how it is administered, estrogen can have a wide range of other health effects. Oral estrogens undergo hepatic first-pass metabolism, which reduces their bioavailability and necessitates greater dosages than their nonoral equivalents. Oral estrogens may negatively impact sexual desire and responsiveness by increasing the hepatic synthesis of sex hormone-binding globulin and decreasing free testosterone. Additionally, oral estrogens increase other hepatic enzymes that affect the vascular, thrombotic, and cardiovascular systems [13].

Transdermal estrogen can be administered as a patch, gel, or spray. Women without a uterus can use estrogen alone. Options for estrogen formulations include micronized 17β-estradiol, CEEs, conjugated estrogens (CE), and ethinyl estradiol. Micronized 17β-estradiol has the same chemical structure as that of estradiol produced by the ovaries; however, the other formulations are synthetic estrogens. The lowest effective dose was recommended when HRT was prescribed. A vaginal ring is available in the form of vaginal estradiol acetate (12.4 mg or 24.8 mg). After 90 days, the vaginal ring is removed and replaced. Low doses of vaginal estrogen can also be used to treat genitourinary symptoms and can be applied as a cream, tablet, insert, or ring [14].

The majority of the main organ systems are affected metabolically by estrogen. Normalizing the vaginal and urethral environment, promoting a non-atherogenic lipid profile, reducing incident diabetes mellitus rates, promoting arterial blood flow and subclinical atherosclerosis, limiting excessive osteoclastic activity, promoting growth, and reducing inflammation contribute to the maintenance of neuronal health in the spine and central nervous system. Individual medications are used more frequently to prevent or treat the consequences of diminished endogenous estrogen levels resulting from the shift away from HRT. For example, statins are used to prevent coronary heart disease. Bone loss is treated with bisphosphonates, selective estrogen receptor modulators, receptor activator of nuclear factor-jB ligand inhibitors, and other methods, whereas hot flushes are treated with selective serotonin reuptake inhibitors. No medication offers the whole range of physiological effects on end organs as estrogen does; each comes with a unique profile of negative effects [7].

The use of estrogen monotherapy beyond age 65 years was associated with significant risk reductions in mortality (19% or adjusted hazards ratio, 0.81; 95% confidence interval, 0.79–0.82), breast cancer (16%), lung cancer (13%), colorectal cancer (12%), congestive heart failure (5%), VTE (3%), atrial fibrillation (4%), acute myocardial infarction (11%), and dementia (2%) [15].

For symptom alleviation, 0.5–1 mg of 17β-estradiol or 0.3–0.45 mg of conjugated equine estrogen is used orally; 2 mg of medium-dose estrogen may also be used to relieve symptoms; and 50 mg of 17β-estradiol or 0.625 mg of conjugated equine estrogen can be used as alternatives [3].

Progestogen therapy

Progestogens are exclusively used in the treatment of menopause to prevent endometrial cancer and hyperplasia. Therefore, progestogens are not recommended for women who have undergone hysterectomy. Significant variations in the metabolic effects and illness outcomes of various progestogens have been discovered in clinical trials, and most observational studies have assessed them [7].

Micronized progesterone, which is bioidentical to endogenous

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progesterone, and synthetic progesterones, such as medroxyprogesterone acetate (MPA, a C-21 derivative) and norethindrone (a C-19 derivative), are examples of progestogens, which are a large class of progestational chemicals [13].

Micronized progesterone and synthetic progestins such as MPA and norenthinrone are examples of progestogen alternatives. Micronized progesterone has effective oral absorption and is bioidentical to endogenously produced hormones. It can also be administered vaginally. Synthetic progestins are 10–100 times more active than are natural progestins. Progestogens can be administered topically or orally. Micronized progesterone dramatically lowers the VMS and promotes better sleep, whereas oral progesterone has moderate sedative effects. The side effects of progestogen treatment include weariness or dizziness, negative mood effects, acne, hirsutism (more prevalent with norethindrone), breast soreness, and edema (more common with MPA) [4].

Compared with other progestogens, micronized progesterone interacts less with androgenic and mineralocorticoid receptors and has a more selective impact on progesterone receptors. According to recent data, HRT regimens that use micronized progesterone may reduce the negative effects and metabolic effects of other progestogens. To minimize adverse progestogenic effects, the progestogen duration can be shortened to 7–10 days, and the progestogen dose can be reduced by half. There should be a low barrier for ultrasound screening and endometrial sampling if clinically warranted because this might lead to bleeding issues and be linked to an increased risk of endometrial hyperplasia [16].

Progestogen may not be administered to women who are uterus-free but is necessary in cases of endometrioid ovarian cancer, if there is a history of endometriosis, or if a portion of the uterus remains after hysterectomy [17].

Combination formulations

Progestins and estrogen are components of combined oral contraceptives (COCs). Since COCs were first introduced in the 1960s, significant advancements and modifications have been made. In 1961, the United States saw the introduction of the first combination contraceptive pill, which comprised 75 μg of mestranol and 5 mg of norethynodrel. Despite being licensed in 1960, a larger dosage of this medication that contained 150 μg of mestranol and 9.85 mg of norethynodrel was never sold as a contraceptive. Studies to determine the lowest effective dose of hormones were not conducted at that time; thus, these high levels were employed because they were determined to be effective. The dosage of sex steroids in COCs has drastically decreased since the initial tablets, allowing for greater safety and fewer side effects. Since the 1960s, research has focused on how estrogen and progestin affect the effectiveness and safety of contraception [18].

The contraceptive action of COCs is mostly mediated by the progestin component. Norethindrone, norethindrone acetate, and ethynodiol diacetate are the first-generation progestins. Second-generation progestins were created in response to the unintended consequences in some women suffering from more unplanned bleeding and spotting as a result of gradual reduction in the dosage of these agents. The second-generation progestins, like levonorgestrel and norgestrel, outperform the first-generation progestins in terms of potency and half-life. These progestin-containing pills have higher androgenic activity, which may increase libido but harm women with dyslipidemia, hirsutism, or acne. Desogestrel and norgestimate, third-generation progestins, were developed to preserve elevated progestational activity while lowering androgenic activity. The estrogen component can manifest itself more completely metabolically when androgenic activity is low. Although this could help with acne, elevated estrogen expression may increase the risk of thromboembolic events, particularly in COCs that include EE. Drospirenone and dienogest, the fourth-generation progestins, are designed to selectively bind to progesterone receptors without interacting with other steroid receptors [18].

Several COC preparations differ in terms of hormone type, dose, and length of hormone-free periods. Monophasic or multiphasic COC formulations with monthly cycling are usually delivered in 28-pill packs. Each active tablet in the monophasic formulation contains the same quantity of progestin and estrogen. A 28-pill box typically consists of 24 or 21 identical active tablets, with the remaining pills serving as placebos. Withdrawal bleeding usually occurs in the absence of placebo hormones. Given that there is a lower chance of folliculogenesis during the hormone-free interval as opposed to 7 days, COCs with 24 active tablets and four placebo pills (24/4) may be more effective. Improved bleeding control and a reduction in hormone withdrawal symptoms such as mood swings, headaches, and pelvic discomfort have also been linked to these formulations. In the multiphasic formulations, the same box of tablets contained several combinations of progestins and estrogens [19]. The formulation of a quadriphasic combination oral contraceptive pill containing estradiol valerate and desogestrel uses a progestogen step-up and estrogen step-down sequence [20].

Over time, several advancements have been made in COC formulation and packaging. The type and quantity of hormones, patterns of these quantities over the course of the cycle, and the quantity of active tablets in the packet can all affect the formulation. The active ingredients in the monophasic formulations contain identical amounts of progestin and estrogen. The quantities of progestin and estrogen in the multiphasic formulations vary. Estrogen and progestin come in two distinct combinations of biphasic formulations and three different combinations of triphasic formulations. A four-phase formulation with a progestin step-up and estrogen step-down sequence was released [18].

Continuous-cyclic estrogen-progestogen, continuous combined estrogen-progestogen treatment (EPT), and intermittent combined EPT are FDA-approved formulations that offer sufficient progestogen dosages for endometrial protection. Daily

estrogen combined with progestogen administered cyclically for 12–14 days every month or every 2–6 months for 14 days in long-cycle formulations is known as continuous-cyclic EPT. Uterine hemorrhage after progestogen withdrawal occurs in up to 80% of women receiving monthly progestogen dosing regimens. While uterine bleeding tends to be thicker and longer, withdrawal bleeding is less common in patients with a prolonged period [4].

The use of continuous COCs or COCs like Qlaira with a shorter pill-free interval is advised when using low-dose COCs since the menopausal symptoms may intensify throughout the 7-day period. If there are no risk factors such as smoking, obesity, high blood pressure, or other cardiovascular disorders, low-dose COCs can be safely administered between the ages of 40 and 55 years. However, it is crucial to assess each individual's risk factors thoroughly [17].

Although there are several alternatives for combination EPT, there is no evidence that one regimen is better than the other. Women with uteruses receiving estrogen therapy alone had a 34% higher incidence of endometrial hyperplasia compared to 1% for those receiving combination therapy, according to the postmenopausal estrogen/progestin intervention study. Comparison of continuous-combined EPT with the general population revealed no elevated risk or potential protection against endometrial cancer [4].

Tibolone is a mixed steroid and 19-nortestosterone derivative that acts biologically through three metabolites. Following consumption, tibolone is converted in the stomach and liver to Δ4-tibolone, an androgen, progestogen, and estrogen metabolite (3α-,3β-hydroxytibolone). It functions similarly to estrogen via the estrogen receptor and has the ability to reduce menopausal symptoms and stop bone loss without damaging the breast or endometrial tissue. This is because it exhibits tissue selectivity owing to changes in enzyme activity. Tibolone is thought to be protective against breast cancer by specifically lowering the levels of estrogen in breast tissue. Although tibolone lacks biological activation, it is classified as a selective tissue estrogen activity regulator because its metabolites exhibit specific therapeutic actions in human tissues [17].

Tissue selective estrogen complex

TSECs combine estrogen and selective estrogen receptor modulators. For usage in postmenopausal women with uteruses, the FDA-approved drug, DuaVee, combines 0.45 mg of oCEE with 20 mg of bazedoxifene, a selective estrogen-receptor modulator. It is used to prevent osteoporosis and treat mild-to-severe VMS. The fact that TSEC did not improve endometrial thickness, breast discomfort, or breast density when compared to that with the placebo may indicate that this treatment is preferable to alternative formulations. More than 83% of the users had amenorrhea. However, its potential to reduce breast cancer risk has not yet been investigated [4].

In addition to treating menopausal symptoms, TSEC is a recently discovered medication that combines CE, bazedoxifene, and selective estrogen receptor modulator to increase the drug tolerance of the current pro-gestationogen and lower the risk of breast cancer, breast discomfort, and vaginal bleeding. Owing to the special pharmacological characteristics and mode of action of bazedoxifene, which acts as an agonist of estrogen receptors in the bone and an antagonist of those in the uterus or breast, combined estrogen/bazedoxifene is used to treat osteoporosis and menopausal symptoms. Furthermore, it ensures endometrial stability while maintaining the benefits of estrogen treatment [17].

Alternatives for Patients with Contraindications for Hormone Replacement Therapy

Some women are medically ineligible for HRT, such as those with a history of venous thrombosis or embolism with a known high risk, absolute contraindication, undiagnosed bleeding in the vagina, liver illness, pregnancy, active CVD, medical history of endometrial or breast cancer, gallbladder illness, uncontrolled hypertension, migraine, and uterine fibroids [8].

Alternatives must be considered because HRT may not be the best option owing to personal preferences or medical contraindications (such as hormone-dependent cancers). In randomized placebo-controlled trials, the prescription of non-hormonal medicines such as paroxetine, fluoxetine, citalopram, escitalopram, venlafaxine, gabapentin, pregabalin, and clonidine were all found to be beneficial. Fezolinetant is an experimental, nonhormonal neurokinin-3 receptor antagonist that is currently being reviewed by the FDA for the treatment of moderate-to-severe VMS. To treat hot flashes and night sweats, kisspeptin, neurokinin B, and dynorphin neurons assist in the regulation of the hypothalamic thermoregulatory regions. Fezolinetant decreased VMS associated with menopause in approximately 60% of the individuals randomized to the therapy group compared to 45% in the placebo group. Paroxetine is a FDA-approved selective serotonin reuptake inhibitor for treating moderate-to-severe postmenopausal VMS at a daily dose of 7.5 mg, which is lower than the level used for psychiatric causes. Other medications with evidence of efficacy in treating VMS include venlafaxine (37.5–150 mg daily), desvenlafaxine (100–150 mg daily), citalopram (10–20 mg daily), and escitalopram (10–20 mg daily). Gabapentin (600–2,400 mg in divided doses) has also been shown to reduce VMS. Clonidine has also been used to treat VMS; however, serious side effects limit its use, including hypotension, rebound hypertension after discontinuation, headaches, lightheadedness, dizziness, dry mouth, constipation, and drowsiness. Oxybutynin, an anticholinergic and antimuscarinic medicine used to treat overactive bladders, has been demonstrated to be beneficial in treating VMS at doses ranging from 2.5 to 5 mg/day extended-release. The Menopause Society

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advises on cognitive-behavioral treatment, clinical hypnotherapy, oxybutynin, weight loss, and stellate ganglion block [14,21].

Risks and Benefits of Hormone Replacement Therapy

In the developed world, menopause in women aged <45 years is classified as premature ovarian failure. Women with premature ovarian insufficiency have an earlier onset of CVD and osteoporosis. They are also noted to have a reduced risk of breast cancer compared to that of their menstruating peers. The risk of breast cancer with HRT use in these women is not deemed to be greater than the population risk for their age, while the benefits are greater due to the prevention of long-term morbidity. Hence, it is strongly advised that these women should consider undergoing HRT until at least 50 years of age [1]. While some guidelines recommend that HRT should not be started after the age of 60 years or 10 years after menopause, evidence supporting this should be reviewed. After 18 years of follow-up, both the Women's Health Initiative (WHI) investigations found no differences in all-cause, CVD, or cancer mortality between the hormone therapy and placebo groups [22].

A meta-analysis of studies assessing the fracture risk in women using oral and transdermal estrogens (with or without the addition of a progestogen) reported a 20%–37% reduced risk of hip, vertebral, and total fractures. HRT in healthy symptomatic women during a naturally timed menopausal transition is associated with symptomatic benefits and low risks. HRT is now recognized in the postmenopausal osteoporosis guidelines. Although not universally approved for bone protection, HRT should be considered as a first option for bone protection in women with early or natural menopause over antiresorptive or osteoanabolic therapies, unless there are over-riding reasons for bone-specific therapy use such as glucocorticoid associated bone loss. HRT may be used indefinitely if its benefits to bone health and menopausal symptoms exceed the risks and adverse effects [22].

Following a 12-week course of therapy, women should be evaluated and their dosage should be modified as needed. HRT needs to be customized, considering each patient's unique circumstances being taken into account. Women who are at high risk of VTE, have migraines, poor liver metabolism, hypertriglyceridemia, or malabsorption disorders, may benefit from the non-oral approach. Some medical professionals advise transdermal estrogens, including micronized progesterone, for women aged >60 years to reduce the risk of venous and arterial thrombosis [3].

Because these recommendations may vary depending on the circumstances of each woman, there is no single safe upper age limit for the use of HRT or an ideal length of therapy. Most women only require short-term therapy to alleviate VMS, whereas others may require long-term HRT. Every woman should be administered the lowest effective dose for the shortest amount of time, and the necessity of continuing HRT should be evaluated at least once a year [1].

To alleviate these uncomfortable effects, HRT should be administered for as short a time as possible. However, some women may experience persistent problems, and stopping treatment may cause undesired symptoms to recur. The woman's preferences and risk/benefit profile should be evaluated at least once a year to determine whether she needs to continue receiving HRT. This can be achieved only by lowering the dosage or by completely stopping the treatment. HRT may be stopped if symptoms do not return, although it should be mentioned that approximately 50% of women report symptom recurrence after stopping HRT [3].

In a 20-year follow-up of the WHI research, women in the estrogen-progestin group had a higher incidence of breast cancer than that of those in the placebo group; however, there was no significant difference in mortality rates. The estrogen-only group showed decreased incidence and mortality rates for breast cancer. The inconsistent evidence on breast cancer risk with estrogen-only therapy could be attributed to a variety of factors, including patient age at HRT initiation, duration since menopause, and screening bias in observational studies. Furthermore, the type of progestogen may affect the risk of breast cancer, as studies have shown that synthetic progestins may be associated with an increased risk of breast cancer compared to that with micronized progesterone [14].

Estrogen is the best therapeutic agent for VMS. For endometrial protection, women with intact uteri must receive both progestogen and estrogen treatments. Estrogen can be used in patients lacking a uterus. When deciding the HRT formulation and route of use, as well as when to discontinue, collaborative decision-making is essential because the risk-benefit profile of HRT treatment in symptomatic menopausal women is influenced by age, duration since menopause, and preexisting comorbidities [4].

In addition to its effects on VMS, HRT also improves depressed mood and prevents tooth loss. Although several studies have found that women who undergo HRT have a lower chance of developing bowel cancer, there is insufficient evidence to recommend HRT as a prophylactic measure. Certain estrogen replacement treatments seem to have neuroprotective properties, maintain cognitive function, and lower the risk of Alzheimer's. There is also evidence of protection against Parkinson's disease. According to certain theories, if HRT is administered early in menopause, there may be a "window of opportunity" to preserve cognitive function. HRT also works well in stopping bone loss, which is typically associated with menopause [1].

According to increasing data, progestogens may alter the estrogen-associated elevated risk of VTE. Those with a history of VTE, such as those with obesity or thrombophilia, are at the highest risk, which seems to be the highest during the first year of usage. To treat unpleasant symptoms, HRT should be administered for as short a time as possible. Nevertheless, some women may experience persistent problems, and discontinuing treatment

may cause symptoms to return in an intolerable manner. Based on women's choices and risk/benefit profiles, the necessity for ongoing HRT should be evaluated at least once a year. This can be achieved only by lowering the dosage or completely stopping the treatment. It should be mentioned that approximately 50% of women report a relapse of symptoms after discontinuing HRT, although if symptoms do not recur, HRT may be stopped [3].

In clinical decision-making during and after the menopausal transition, it is essential to comprehend and keep an eye on the dangers associated with HRT. Age, lifestyle changes, and menopause-related symptoms (vasomotor, sleep disruption, vaginal atrophy, and dyspareunia) all affect a woman's benefit/risk ratio [3].

The risk of endometrial cancer (EC) is closely linked to circulating estrogen and androgen levels [13], and unchecked estrogen use increases the risk of endometrial hyperplasia [12]. In contrast, progestogens are thought to reduce the number of endometrial tumors and prevent carcinogenesis. According to the WHI study, the risk of EC is considerably reduced by daily use of a synthetic progestin combined with estradiol for 5 years. Furthermore, there may be an interaction between body mass index and HRT in EC. With an additional risk of 1.59 per 5 kg/m^2, the risk of obesity-related EC may be specifically explained by a hyperestrogenic condition caused by increased rates of androgenic precursor conversion to estradiol in the adipose tissue. This condition of high estrogen is experienced by postmenopausal women taking HRT [23].

Conclusion

The advantages and hazards of hormonal treatment in postmenopausal women are influenced by age and the amount of time following menopause. Estrogen medication should typically be continued until the average age of natural menopause in women with premature or early onset menopause. HRT offers good symptom alleviation and poses a minimal risk for healthy women with menopausal symptoms at the typical age of natural menopause. Women who exhibit symptoms may be at risk if HRT is withheld, especially if they have osteoporosis and CVDs. Conversely, HRT is often avoided and may be associated with a higher risk in older women. In order to maximize treatment for women with menopausal problems, the abundance of clinical trial data in recent years not only permits but also screams for the customization of hormone therapy decision-making, even though it might occasionally be intimidating to the prescribing practitioner.

It is necessary to have customized conversations about the relative advantages and disadvantages of HRT for women with menopausal symptoms, considering their unique health situation. Nonhormonal options such as herbal supplements, mind-body practices, lifestyle changes, and nonhormonal medications should also be explained to patients. Considerations should be made regarding the woman's age, the type and timing of menopause, the effect of symptoms on quality of life, her medical history, her family's medical history, and her personal preferences.

Estrogen, with or without progestogen, is the most effective therapy for menopausal symptoms, such as urogenital atrophy, night sweats, and hot flashes. HRT is most helpful before the age of 60 years or within 10 years of menopause. HRT may alleviate mood swings, sleep issues, sexual dysfunction, and myalgia, among other menopausal symptoms. HRT is also useful for preventing menopausal bone loss and lowering the risk of osteoporosis-related fractures, including hip and vertebral fractures. For postmenopausal women aged <60 years who are at a higher risk of fracture, HRT may be one of the first-line treatments for osteoporosis prevention and treatment. When HRT is stopped, its protective impact on bone mineral density diminishes.

Attempts should be made to decrease and finally stop hormone therapy after 3–5 years. Lower doses or transdermal treatment may be recommended if symptoms continue, with the risks and advantages being reassessed periodically. Low-dose vaginal hormone therapy may be available if vaginal moisturizers and lubricants are insufficient to address genitourinary symptoms after medication termination.

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Conceptualization: AC. Methodology: ME, ET. Investigation: ME, ET. Validation: AC, VA. Supervision: VA. Writing–original draft: ME, AC. Writing–review & editing: ET, VA. Final approval of the manuscript: all authors.

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